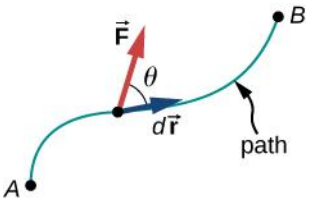
Chapter 38 - Electric Potential Energy and the Electric Potential
Calculus Physics II

Work – Work in Physics is the amount of energy required to change the position or motion of an object by applying a force ($\vec{\mathbf{F}}$) over some displacement ($d\vec{\mathbf{x}}$). Work is the integral of the dot product of the force and incremental displacement vectors ($d\vec{r}$).

$$W = \int_A^B \vec{F} \bullet d\vec{r} = \int_A^B |\vec{F}| \cos(\Theta) dr$$



Units of Work/Energy are Joules (1 Nm = 1 J)

Mechanical Energy – Mechanical Energy has the ability to do work on an object and changes that object, the energy can be either potential (PE) or kinetic (KE).

Kinetic Energy – The energy of motion; $KE_{translation} = \frac{1}{2}mv^2$ or $KE_{rotational} = \frac{1}{2}I\omega^2$.

Potential Energy - is the ability to do work by virtue of the objects position or stored energy in a field.

Similarities between PE_g and PE_E - Because the Gravitational force and the Electrostatic force are conserved, they both have potential energies, the difference in the potential energy is equal to the (negative) work done by the field. It is the change in potential energy that matters because the "zero" point for the potential energy of an object can be defined anywhere.

Gravitational Potential Energy: $PE_g = m|\vec{g}||\Delta\vec{y}|$ ($\vec{g}$ is constant over $\Delta\vec{y}$) $\Delta PE_g = -W_{g\text{-field}}$

***Electrical Potential Energy*:** $PE_E = |\vec{F}_E||\Delta\vec{x}| = q|\vec{E}||\Delta\vec{x}|$ (constant $\vec{E}$ over $\Delta\vec{x}$) $\Delta PE_E = -W_{E\text{-field}}$

- The gravitational Potential Energy exists because there is a gravitational field.
- The Electric Potential Energy exists because there is an Electric field.
- The gravitational field occupies all of space and is created by the mass.
- The Electric field occupies all of space and is created by an object with a net charge.

Work and the change in Electric Potential energy (PE_E or U_E): The amount of work done on an object changes the potential energy of that object in a conservative field. The work done on an object is equal to the negative change in potential energy of that object. Work done by a force acting over a distance from point A to point B is given in general as

$$W = \int_A^B \vec{F} \bullet d\vec{x} = -\Delta PE$$

The work done by the force is determined by taking the dot product of the force vector with the incremental displacement integrated over the change in the position of the object from position A to position B.

- The applied force does maximum positive work when the force and displacement are in the same direction.
- The force does no work when the force and displacement are perpendicular to each other.
- The force does negative when the angle between the force and displacement is obtuse.

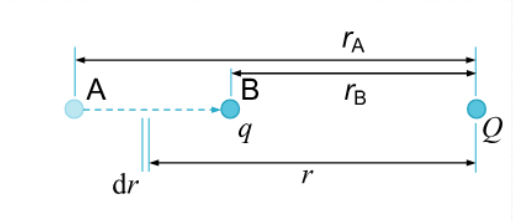
Electric potential energy is the energy a charge possesses due to its position in an electric field. It is the work required to move a charge from a reference point to its current location against the electric force.

- For the same type of point charge (both positive or both negative) – Their potential energy increases as they move closer together and their potential energy decreases as they move farther apart.

- For two different type of point charge (one is positive and the other negative) – There potential energy decreases as the move closer together and their potential energy increases as they move farther apart.

For two point charges Q and q, assuming the point charge Q is fixed and point charge q moves from point A to B. The force between two point charges is given as

$$\vec{F}_E = \left[k\frac{q_1 q_2}{r^2}\right]\hat{r}$$



Assuming the radius r is along the x-axis (x = r), the work done to move the second point charge q is given as

$$W = \int_A^B \vec{F} \bullet d\vec{x} = \int_A^B \left[k\frac{Qq}{r^2}\right]\hat{r} \bullet d\vec{x} = kqQ\int_A^B \left[\frac{1}{r^2}\right]\cos(\varphi)dr = -kqQ\cos(\varphi)\left[\frac{1}{r_B} - \frac{1}{r_A}\right]$$

Where the angle phi φ is between the force vector and the displacement vector. For the force and displacement in the same direction φ = 0° and cos(0°) = 1. For the force and displacement vectors in opposite directions φ = 180° and cos(180°) = -1. Before determining the integral always determine the relative directions of the force and displacement vector and apply the appropriate sign.

Given that the potential energy is equal to the negative of the work done

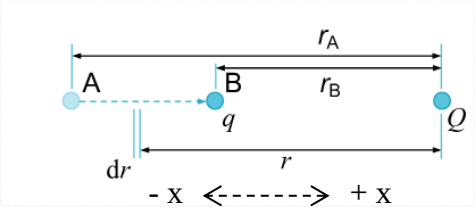
$$W = -\Delta PE_E = = -kqQ\cos(\varphi)\left[\frac{1}{r_B} - \frac{1}{r_A}\right]$$

The change in electrical potential energy of point charge q moving from point B to point A is given as

$$\Delta PE_E = kqQ\left[\frac{1}{r_B} - \frac{1}{r_A}\right]$$

- The change in electrical potential energy between two point charges depend on the relative change in distance between the charges and the type of charges that are interacting. The relative direction between the force and displacement vector is determined by the types of charge that are interacting, this will determine the angle φ and indicate if energy is lost or gained by the object.

Example: The point charge Q has a net charge of 3 nC (3 x 10^{-9} C) and is positively charged, the point charge q has a net charge of 5 nC (5 x 10^{-9} C) and is positively charged. The radius from Q to B is equal to r_B = 10 cm and to A is equal to r_A = 15 cm. (k = 9.0 x 10^{9} Nm^2/C^2)



A) Determine the change in electric potential energy of point charge q moving from point B to point A.

$$\Delta PE_E = -\int_B^A \left[k\frac{Qq}{r^2}\right]\hat{r} \bullet d\vec{r} = -kqQ\int_B^A \left[\frac{1}{r^2}\right]dr = -kqQ(1)\left[\frac{1}{r_A} - \frac{1}{r_B}\right]$$

$$\Delta PE_E = -4.5\ x\ 10^{-7} J$$

- Both charges are positive therefore the force that q experience due to the electric force is in the negative x direction and the displacement is in the negative x direction φ = 0°. The point charge q gains kinetic energy but loses electric potential energy.

B) Determine the change in electric potential energy of point charge q moving from point A back to point B.

$$\Delta PE_E = -\int_A^B \left[k\frac{Qq}{r^2}\right]\hat{r} \bullet d\vec{x} = -kqQ\int_A^B \left[\frac{1}{r^2}\right]dr = -kqQ\left[\frac{1}{r_B} - \frac{1}{r_A}\right]$$

$$\Delta PE_E = +4.5\ x\ 10^{-7} J$$

- The electric force that q experience at A is in the negative x direction but q's displacement is in the positive x direction φ = 180°. The point charge q gains potential energy going from B to A.

C) Let's now assume that the point charge Q is a negative charge but has the same value of 3 nC. Determine the change in electric potential energy of point charge q moving from point A to point B.

$$\Delta PE_E = -\int_B^A \left[k\frac{Qq}{r^2}\right]\hat{r} \bullet d\vec{x} = kqQ\int_B^A \left[\frac{1}{r^2}\right]d\vec{r} = kqQ\left[\frac{1}{r_A} - \frac{1}{r_B}\right] = 4.5\ x\ 10^{-7} J$$

- The point charge q experiences a electric force in the positive x direction (attraction) at B. Charge q's displacement is in the negative x direction therefore φ = 180°. The point charge q gains potential energy going from B to A.

Zero electric potential energy - The electric potential energy of a charge can be defined as zero at infinity because it represents the state where the charge is infinitely far from any other charges and thus experiences no electric force or interaction. At infinity, there is no potential energy stored due to electrostatic interactions, there is no electric field.

Reference Point: The choice of where to set the zero point for potential energy is arbitrary, but for point charges and localized charge distributions, it is convenient to set it at infinity.

Defining the potential energy as zero at infinity simplifies calculations and provides a consistent reference point for potential energy calculations involving point charges and other localized charge distributions. It's important to note that the zero potential at infinity is a convention, not a physical law. It's a choice made for mathematics and convenience.

$$PE_E = kqQ\left[\frac{1}{r_B} - \frac{1}{\infty}\right] = kqQ\left[\frac{1}{r}\right]$$

Where r is the distance between the two charged object.

When assuming the electric potential energy of the system is zero at infinity the above equation gives the energy of the system of charged particles a distance r apart. It does not determine the energy of just one of the particles.

Example: Using the point charges Q and q from above, determine the potential energy of the system

A) At point B. r_B = 10 cm

$$At\ point\ B \quad PE_E = kqQ\left[\frac{1}{r}\right] = 1.35\ x\ 10^{-6}\ J$$

B) At point A. r_A = 15 cm

$$At\ point\ A\quad PE_E = kqQ\left[\frac{1}{r}\right] = 9.0\ x\ 10^{-7}\ J$$

C) Determine the difference in potential energies between point B and point A.

$$\Delta PE_E = 1.35\ x\ 10^{-6}\ J - 9.0\ x\ 10^{-7}\ J = 4.5\ x\ 10^{-7}\ J$$

- This is the same change in electric potential energy as in the above example the change in the electric potential energy has not changed.

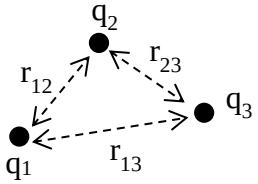
The total potential energy of a system of charged particles is given as the work required to bring each charged particle to a point in space from infinity. The total systems electric potential energy for a system of point charges is the potential energy gained (work done on the particle) by bringing each particle from infinity to a fixed location in space.

$$PE_{system} = PE_{1st} + PE_{2nd} + PE_{3rd} + \cdots$$

Example: Given 3 positive point charges with charge of q_1 = 3.1 μC, q_2 = 2.5 μC and q_3 = 2.0 μC. The separation of the charges is shown r_{12} = 17 cm, r_{23} = 20 cm and r_{13} = 25 cm. Determine the total electric potential energy of the system of charges.

Bring the first charge q_1 from infinity to empty space requires no work and therefore has no electric potential energy by itself.

$$PE_{1st} = 0\ \text{J}$$



Bring the second charge q_2 to r_{12} will require work between q_1 and q_2.

$$PE_{2nd} = k\left[\frac{q_1 q_2}{r_{12}}\right] = 0.41\ J$$

Moving q_3 to position r_{13} requires work against both q_1 and then q_2

$$PE_{3rd} = k\left[\frac{q_2 q_3}{r_{23}}\right] + k\left[\frac{q_1 q_3}{r_{13}}\right] = 0.225\ J + 0.223\ J = 0.448\ J$$

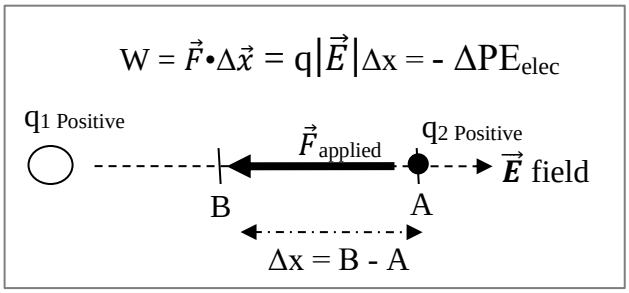
The total potential energy of the system of charged particles is given as

$$PE_{system} = PE_{1st} + PE_{2nd} + PE_{3rd} = 0\ J + 0.41\ J + 0.448\ J = 0.86\ J$$

<u>Work and the Electric Potential Energy in a uniform electric field</u>– The work done by a uniform electric field on a charged particle is directly related to the change in its electric potential energy. This relationship is essential for understanding the energy transformations that occur as charged particles move within electric fields.

An applied force ($\vec{F}_{\text{applied}}$) over a distance ($\Delta\vec{x}$) or Work ($W = |\vec{\mathbf{F}}||\mathbf{\Delta\vec{x}}|\cos\varphi$) is required to move a positive charged object closer to a second fixed positive charged object or to separate a positive charged object from a negative charged object in a uniform electric field.

- The work done on an object with a net charge must be equal to the change it the potential energy gained by the charged object if the force is a conservative force.



<u>For a uniform Field</u> – When a charged object is in a uniform electric field it experiences a force $\vec{F}_E = q\vec{E}$ Therefore

$$W = \int_{r_A}^{r_B} \vec{F} \bullet d\vec{x} = \int_{r_A}^{r_B} (q\vec{E}) \bullet d\vec{x} = q|\vec{E}|\cos(\varphi)\int_{r_A}^{r_B} dx = q|\vec{E}|\cos(\varphi)\,(r_B - r_A)$$

$$W = q|\vec{\mathbf{E}}||\mathbf{\Delta\vec{x}}|\cos\varphi$$

- If the Electric field is doing the work on a charged object, then the electric field and the displacement are in the same direction and the work is positive.
- If an applied force is doing the work on the charged object, then the Electric field is in the opposite direction as the displacement and the work done is negative.
- Work on an object is only done by the component of the force parallel to the electric field lines. No work is done on a charged object moving perpendicular to the electric field lines.

The work done on the charged object by the electric force creates the potential energy the object gains or is lost as the object moves along the Electric field lines.

$$\Delta PE_E = -W = -\,q|\vec{\mathbf{E}}||\mathbf{\Delta\vec{x}}|\cos\varphi$$

Electric potential (V) - is a measure of the potential energy per unit charge at a specific point in an electric field. It indicates the amount of work needed to move a unit positive charge from a reference point (often infinity) to that specific point.

- The standard unit for electric potential is the Volt (V), which is equivalent to Joules per coulomb (J/C).
- Electric potential is also known as voltage.
- The Electric Potential is defined by the positive test charge because the electric field is defined by the positive test charge!

$$Potential\ V = \frac{Electic\ Potential\ Energy\ of\ the\ charged\ object\ in\ an\ Electric\ field}{Total\ amount\ of\ net\ charge\ on\ the\ object}$$

Units are Volts (V) (1 V = 1 J/C) amount of work per unit charge

- Electric potential is a scalar! No direction just magnitude.
- Electric potential is the potential energy that a positive unit charge would possess at a given location in an electric field.
- The electric potential at infinity is typically taken as zero, allowing for the calculation of potential at other points.
- Electric potential is defined by the electric field. The electric field is the negative gradient of the electric potential.

The electric potential similar to the electric potential energy is a function of the position in the electric field. The difference is that the electric potential is defined by a positive test charge and does not change with type of charge. The electric potential energy will decrease in the direction of the electric field for a positive charge but increase for a negative charge.

- If a positive charge moves from high potential to low potential, the charge moves through a potential difference and moves in the direction of the electric field.

  Note: A neutral particle would have no Electric potential energy in an electric field therefore the potential difference would not have an effect on the neutral particle.

Potential (ΔV) and Electric Potential Energy (PE_E) – They are not the same thing! The Potential Energy PE_E is the ability of the charged particle to do work because of where it is in the Electric field and has units of Joules. The Potential V at a point in the Electric field is the amount of energy a charge would have if placed there due to the work required to move the charge to that point and has units of Volts.

$$\Delta V = \frac{\Delta PE_E}{q} \quad or \quad \Delta V = \frac{-W}{q} \qquad\qquad \Delta PE_E = q\Delta V$$

Work done by the field is equal to the change in Electric potential over a distance times the charge that is being moved.

$$W_{AB} = - \Delta PE_{elec} = q(V_A - V_B) = - q\Delta V$$

$$\Delta PE_E = q\Delta V$$

◊ The electric potential difference (ΔV) depends on the electric field and how it is changing over a distance. It is another way to view the electric field and the electric force in terms of energy.

- The negative sign indicates that the potential difference decreases along the electric field line.

$$\boxed{\Delta V = \Delta PE_E / q = -W/q}$$

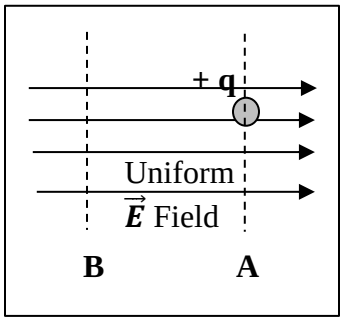
Example: The potential difference ΔV between A and B is 200 V and is separated by 0.5 m, how much work is required to move a positive 2 μC charged sphere from position A to B in an electric field (E = 400 V/m)?

$$W = F_E \Delta x \cos\varphi = (qE)\Delta x \cos\varphi = (2 \times 10^{-6}\ C)(400\ V/m)(0.5\ m) = 4 \times 10^{-4}\ J$$

Where $\varphi = 0°$ along the $\vec{E}$ field

What is the potential energy of the 2 μC charged object at position B in the electric field relative to position A?

$$\Delta PE_E = q\Delta V = (2 \times 10^{-6}\ C)(200\ V) = 4 \times 10^{-4}\ J$$



Electric Field ($\vec{E}$) and the Electric Potential (V) - The electric field is a vector quantity representing the force per unit charge, while the electric potential is a scalar quantity representing the potential energy per unit charge.

The electric field is defined as the negative gradient of the electric potential. This means that the electric field points in the direction of the decrease in electric potential. Mathematically, this relationship is expressed as:

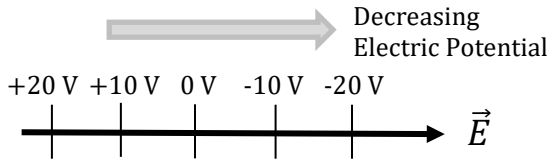
$$\vec{E} = -\nabla V = -\left[\left(\frac{\partial V}{\partial x}\right)\hat{x} + \left(\frac{\partial V}{\partial y}\right)\hat{y} + \left(\frac{\partial V}{\partial z}\right)\hat{z}\right]$$

The gradient (∇) is a vector operator that acts on a scalar field and produces a vector field. It indicates the direction and magnitude of the slope of the scalar field. The mathematical expression means that the gradient of V is a vector whose components are the rates of change of V along the x, y, and z directions. The resulting vector points in the direction where V increases most rapidly, and its magnitude represents the rate of that change.

In one-dimensional space the gradient of the electric field is given as

$$\vec{E} = -\left(\frac{dV}{dx}\right)\hat{x} \qquad E_x = -\frac{dV}{dx}$$

- This represents the rate and direction of the change in electric potential.
- Negative Sign: The electric field points in the direction of decreasing potential. So, if potential increases in one direction, the electric field points the opposite way.



- The stronger the electric field, the larger the change in the potentials change with position.
- The electric field at a point depends only on how the potential is changing at that point, not on the absolute value of the potential.

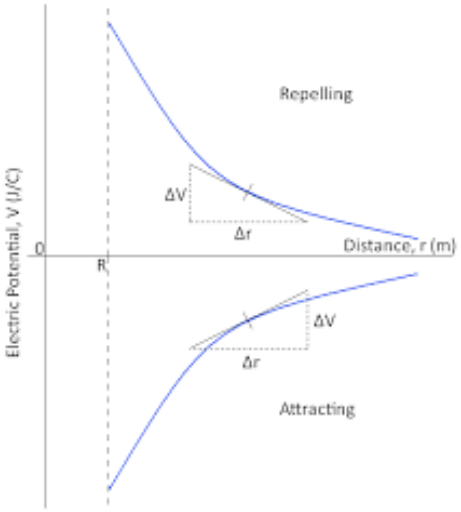
<u>Graphing Electric Potential (V) vs. displacement (Δx)</u> – for a graph of the potential vs. the displacement, if the relationship is linear then the slope of the line is the magnitude of the electric field.

$$\left|\vec{E}\right| = -\frac{dV}{dx}$$

- Tangent line to the curve gives the magnitude of the electric field.
- Constant slope means a constant $\vec{E}$ over the displacement of the charge.
- Units of the slope are V/m which means the units of the electric field are also V/m.

For the same type of charges (top graph - repelling).

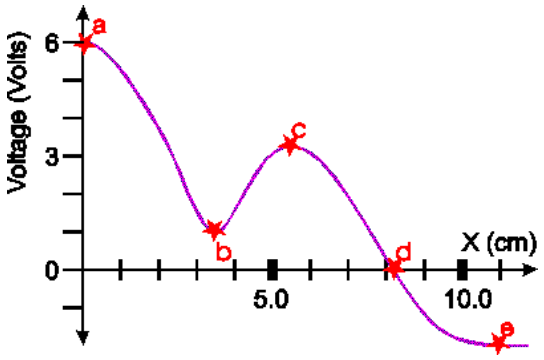
- ❖ Negative slope of the tangent line means the electric field points along the radius R going away from the charge.
- ❖ The magnitude of the slope of the tangent line decreases with increasing R, therefore the magnitude of the electric field decreases with increasing R.



For dissimilar charges (bottom graph - attraction).

- ❖ Positive slope of the tangent line means the electric field points towards the charged object.
- ❖ The magnitude of the slope of the tangent line decreases with increasing R, therefore the magnitude of the electric field decreases with increasing R.

Example: Given the graph of electric potential vs displacement below.



Positions a, b, and c – the electric field is zero because the slope of the tangent line is zero.

Between a and b – the slope is negative and nearly constant therefore the electric field is directed in the positive x direction and is nearly constant in this region.

Between b and c – the slope is positive therefore the electric field is directed in the negative x direction.

Between c and e – the slope is again negative therefore the electric field is pointing in the positive x direction and at point d the electric potential is zero but there is an electric field (there is a slope to the line at d).

- Note the magnitude of the electric fields between a and b, b and c, c and e would be determined by the slope of the tangent line at a point on the curve in that region,

Example: An electric potential along a line is given as

$$V(x) = Ax^2 + Bx + C$$

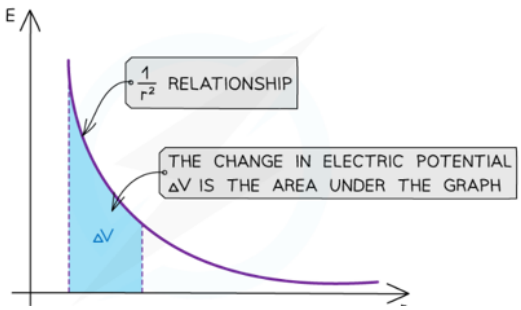
Determine the equation for the electric field along the x axis

$$\vec{E} = -\left(\frac{dV}{dx}\right)\hat{x} = = -\left(\frac{d}{dx}(Ax^2 + Bx + C)\right)\hat{x} = -(2ax + B)\hat{x}$$

- Do not forget the unit vector, the electric field is a vector!

Conversely, the electric potential difference between two points can be determined by integrating the electric field along a path connecting those points. The potential difference along a given axis can be determined by

$$\Delta V = V_B - V_B = -\int_A^B \vec{E} \bullet d\vec{r}$$



Example: Given the electric field vector below, determine the equation for the changing electric potential from 0 to d.

$$\vec{E} = (De^{-mx})\hat{x}$$

$$\Delta V = -\int_A^B \vec{E} \bullet d\vec{x} = -\int_0^d (De^{-mx})\, dx = -\left[-\frac{D}{m}e^{-mx}\right]\frac{d}{0} = \left[\frac{D}{m}e^{-md} + \frac{D}{m}\right]$$

<u>Electric Potential & a Uniform Electric Field</u> – Assuming that in a certain region the electric field is uniform (the same magnitude) and is pointed in the positive x-direction with a magnitude E_x. Because qE_x is the force on a unit positive test charge, the work done in moving the test charge through a distance Δx is W = $\vec{\mathbf{F}} \bullet \Delta\vec{\mathbf{x}}$ and $\vec{F}_E = q\vec{E}$,

$$\Delta V = -\vec{\mathbf{E}} \bullet \Delta\vec{\mathbf{x}} = -\left|\vec{\mathbf{E}}\right||\Delta\vec{\mathbf{x}}|\, cos(\phi)$$

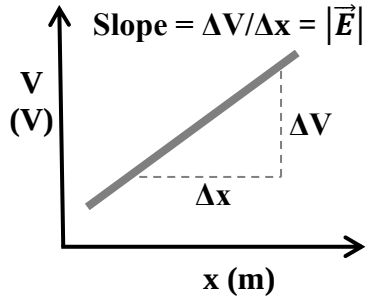
The angle ϕ is between the two vectors $\vec{\mathbf{E}}$ and $\Delta\vec{\mathbf{x}}$. For an object moving along a straight line (x-axis) then (φ is 0° or 180°).

The change in the electric potential is proportional to the magnitude of the electric field.

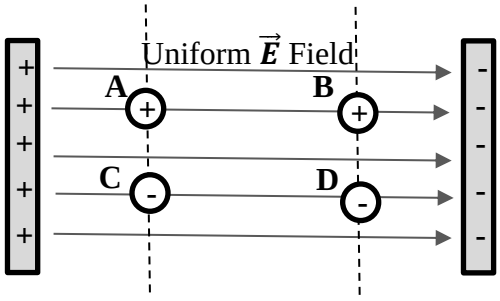
$$\left|\vec{E}\right| = -\frac{dV}{dx}$$

For a constant electric field, the magnitude of the field is equal to the change in the electric potential over the change in the displacement along the field.

$$\left|\vec{E}\right| = -\frac{\Delta V}{\Delta x}$$



<u>Example:</u> Four equally charged spheres are placed in a uniform electric field as shown, Spheres A and B have a positive charge and Spheres C and D have a negative charge.



a) Between spheres A and B which has the large potential energy, and which is at the larger potential?

Sphere A has the larger potential energy because it has a positive charge and is farther up in the electric field than sphere B. Sphere A is closer to the positive plate therefore it has more PE. Sphere A is at the larger potential because potential is farther up the electric field and closer to the positive plate. Potential decreases in the direction of the electric field.

b) Between C and D which has the larger potential energy, and which is at the larger potential?

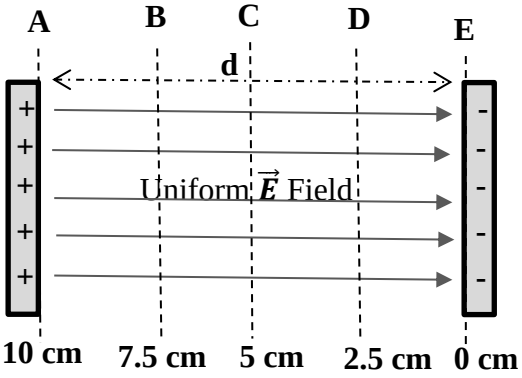
Sphere D has the larger potential energy because it is a negative charge that is father down the electric field, and it is closer to the negative plate. Sphere C is at the larger potential because the potential is defined by the positive charge and decreases in the direction of the electric field.

c) If the spheres are equally spaced for each plate, then sphere A has the same potential energy as sphere D and sphere C has the same potential energy as B.

Example: Two charged parallel conducting plates are placed d = 10 cm apart, if the electric field is uniform and 2000 V/m,

a) What is the potential difference between the plates?

$\Delta V = - E_x d = (2000\ V/m)(0.1\ m) = - 200\ V$



The change in electric potential is negative because the potential is decreasing from Point A to Point E.

b) What is the potential at A, B, C, D and E inside the plates?

Position A

$\Delta V_{A-E} = - E_x \Delta x_A = (2000\ V/m)(0.1\ m) = - 200\ V$

Position B

$\Delta V_{B-E} = - E_x \Delta x_B = (2000\ V/m)(0.075\ m) = -150\ V$

Position C

$\Delta V_{C-E} = - E_x \Delta x_C = (2000\ V/m)(0.05\ m) = -100\ V$

Position D

$\Delta V_{D-E} = - E_x \Delta x_D = (2000\ V/m)(0.025\ m) = -50\ V$

Position E

$$\Delta V_{E\text{-}E} = - E_x \Delta x_E = (2000 \text{ V/m})(0 \text{ m}) = 0 \text{ V}$$

c) What is the potential difference between each position?

$\Delta V_{A\text{-}B} = V_B - V_A = - 150 \text{ V} - (-200 \text{ V}) = 50 \text{ V}$
$\Delta V_{B\text{-}C} = V_C - V_B = - 100 \text{ V} - (-150 \text{ V}) = 50 \text{ V}$
$\Delta V_{C\text{-}D} = V_D - V_C = - 50 \text{ V} - (-100 \text{ V}) = 50 \text{ V}$
$\Delta V_{D\text{-}E} = V_E - V_D = 0 \text{ V} - (-50 \text{ V}) = 50 \text{ V}$

For a uniform (constant) electric field, the change in the electric potential for a given change in position is constant, $\Delta V = E/\Delta x$.

d) If a charged sphere (q = 10 µC positive) where placed between the plates along the B line what potential energy would the charged sphere have?

$$PE_E = q\Delta V = (10 \text{ x } 10^{-6} \text{ C})(150 \text{ V}) = 0.0015 \text{ J}$$